The One-Neutralino Universe: Simulating Benevolent Structural Identity

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Neutralino $\tilde{\chi}_1^0$ (SUSY BSM) | Spin: 1/2 | Mass: ~100 GeV–1 TeV | Majorana fermion | Lightest supersymmetric particle **Source:** Extends `v1.0Deep Dive_ Benevolent Neutralinos.pdf` (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
LHC Run 3 SUSY limits	Pre-Run 3 mass limits	V2.0: ATLAS/CMS Run 3 (2022-2025): no neutralino signal. Simplified model exclusion: $m_{\tilde{\chi}_1^0}$ > 900 GeV (wino-like), > 1100 GeV (higgsino, compressed spectrum). V2.0 simulation mass range updated to 0.9-5 TeV	ATLAS SUSY summary 2024; CMS SUSY Paper 2024
Direct detection exclusion	General WIMP sensitivity	LZ 2024 constrains all neutralino thermal relics at m_χ < 900 GeV (for weak-scale coupling). V2.0: Neutralino must have mass > 1 TeV OR non-thermal production to remain viable dark matter candidate	LZ 2024 PRL

Parameter	V1.0	V2.0	Source
Majorana fermion simulation	Conceptual description	V2.0: Majorana condition ψ = ψ^c implemented as: particle and antiparticle are the SAME worldline segment. In OEU framework, neutralino has no distinct antiparticle — every encounter is self-annihilation candidate. Data structure: bool IS_MAJORANA = true — no separate antiparticle registry	Majorana 1937; Modern context: Wilczek 2024 review
Funnel region annihilation	General SUSY	V2.0: A-funnel (m_A ~ 2*m_χ̃) and Z/h-funnel regions now precisely mapped from LHC Higgs data. Neutralino annihilation rate 10×-100× enhanced in funnel — creates spectacular visual eruption event when two worldlines are in funnel band	MSSM H/A funnel 2024 ATLAS

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1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the "Origami Universe" metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,          // Spinor components
    berry_phase: f32,     // Geometric phase accumulated
```

```
    proper_time: f32,     // tau - worldline parameter
    color_charge: u32,    // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32, // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}
```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m_e	9.10938356×10^{-31} kg	$9.1093837139\times10^{-31}$ kg
e	1.60217663×10^{-19} C	$1.602176634\times10^{-19}$ C (exact)
hbar	$1.054571817\times10^{-34}$ J·s	$1.054571817\times10^{-34}$ J·s (exact)
c	2.99792458×10^{8} m/s	2.99792458×10^{8} m/s (exact)
alpha	1/137.036	1/137.035999084 ±0.000000021

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2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```
// V2.0 Universal Particle State — extends V1.0
struct ParticleInstanceV2 {
    // ── Inherited from V1.0 ─────────────────────────────
    double proper_time;          // tau worldline parameter (meter)
    glm::dvec3 position;         // 3-space vector
    glm::dvec3 momentum;         // 3-momentum (relativistic)
```

```
    float spin_z;                 // +0.5 or -0.5 (or eigenvalue)
    float berry_phase;            // Accumulated Berry/Aharonov-Bohm phase

    // □□ V2.0 ADDITIONS □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    float entanglement_r;         // Reduced density matrix purity 0-1
    int entanglement_partner_id;  // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
    uint32_t color_bits;          // 3-bit: R(b0),G(b1),B(b2)
    float decay_probability_dt;   // P(decay in dt) for unstable particles
    float vacuum_alignment;       // Order parameter for condensate mode

    // □□ WebGPU Buffer packing □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // Total: ~100 bytes/particle □ 10^6 particles = 100 MB GPU memory
    float _pad[2];                // Align to 16 bytes for WGSL
};
```

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8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = **Quantum Coherence** + **Information Preservation** + **Symmetry Maintenance**

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - C(t) = l_1-norm coherence of particle density matrix (Baumgratz et al. 2014) - F(t) = quantum Fisher information / classical Fisher information (ratio □ 1) - S(t) = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

End of v1.0Deep Dive_ Benevolent Neutralinos V2.0 | Tardigradia Game Engine SubParticles Series This document supersedes V1.0 in all physics specifications.